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1070 REM SKIP EARTH
1070 IF I = 3 THEN NEXT
1080 GOSUB 1370
1090 R(I) = Q: X(I) = X: R(I) = R: V(I) = V
1100 NEXT
1110 FOR K = 1 TO 2
1120 I = K: A(I) = FN DEG(A(I))
1130 GOSUB 1710
1140 NEXT
1150 PRINT : PRINT : PRINT
1160 INPUT "WANT ANOTHER DATE Y/N? "; A$
1170 IF A$ = "Y" THEN GOTO 380
1180 IF A$ < > "N" THEN PRINT "INVALID RESPONSE": PRINT : GOTO 1160
1190 HOME
1200 END
1210 REM CALC POSITION DATA
1220 REM CALC HELIOCENTRIC LONGITUDE
1230 R = NI * PD(J,0) + PD(J,1)
1240 IF A > 6.28318 THEN A = ((A / 6.28318) - INT (A / 6.28318)) * 6.28318
1250 IF A < 0 THEN A = A + 6.28318: GOTO 1250
1260 C = PD(J,2) * SIN (A - PD(J,3))
1270 A = A + C
1280 IF A > 6.28318 THEN A = A - 6.28318
1290 IF A < 0 THEN A = A + 6.28318: GOTO 1290
1300 REM CALC PLANET DIST FROM SUN
1310 DS = PD(J,4) + PD(J,5) * SIN (A - PD(J,6))
1320 REM CALC DIST FROM ECLIPTIC
1330 L = PD(J,7) * SIN (A - PD(J,8))
1340 RETURN
1350 REM CALC PLANETARY DATA
1360 REM CALC ANG DISTANCE FROM EARTH
1370 Z = A(3) - A(I)
1380 IF ABS (Z) > 3.14159 AND Z < 0 THEN Z = Z + 6.28318
1390 IF ABS (Z) > 3.14159 AND Z > 0 THEN Z = Z - 6.28318
1400 REM CALC DISTANCE FROM EARTH
1410 Q = SQR (D(I) ^ 2 + D(3) ^ 2 - 2 * D(I) * D(3) * COS (Z))
1420 REM CALC ANG DISTANCE FROM SUN
1430 P = (D(I) + D(3) + Q) / 2
1440 X = Z + FN ACO( SQR ((P * (P - D(I)) / (D(3) + Q))) )
1450 REM CALC RA OF PLANET
1460 IF Z < 0 THEN R = FN DEG(A(3) + 3.14159 - X) / 15
1470 IF Z > 0 THEN R = FN DEG(A(3) + 3.14159 + X) / 15
1480 IF R > 24 THEN R = R - 24: GOTO 1480
1490 IF R < -24 THEN R = R + 24: GOTO 1490
1500 IF R < 0 THEN R = R + 24: GOTO 1500
1510 REM CALC DECLINATION
1520 IF Z < 0 THEN V = SIN (A(3) + 3.14159 - X) * 23.44194 + FN DEG(L(I))
1530 IF Z > 0 THEN V = SIN (A(3) + 3.14159 + X) * 23.44194 + FN DEG(L(I))
1540 X = FN DEG(X)
1550 R(3) = FN DEG(A(3) + 3.14159) / 15
1560 IF R(3) > 24 THEN R(3) = R(3) - 24: GOTO 1560
1570 IF R(3) < -24 THEN R(3) = R(3) + 24: GOTO 1570
1580 IF R(3) < 0 THEN R(3) = R(3) + 24
1590 V(3) = SIN (A(3) + 3.14159) * 23.44194
1600 RETURN

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